CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 2



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Recap of Previous Lecture



Topic Pipeline Processing

Topic Pipeline Cycle Time

Topic Speed Up

Topics to be Covered



- **Topic** Pipeline Throughput
- **Topic** Latency
- **Topic** Instruction Pipeline



Topic : Latency and Throughput

non-pipeline $\Rightarrow t_n$

pipeline $\Rightarrow t_p$

Latency :-

After how much time a new input is given.

Throughput = no. of operations performed per unit time

Non-pipeline :-

in t_n time no. of operations = 1

in 1 time —— 11 —— $= \frac{1}{t_n}$

pipeline :-

in $(k+n-1)\,t_p$ time no. of operations = n

in 1 —— 11 —— $= \frac{n}{(k+n-1)\,t_p}$

in ideal condition Throughput $= \frac{1}{t_p}$

[NAT]

#Q. Consider 6 segment pipeline with segment delay of segments as 120ns, 126ns, 121ns, 110ns, 118ns and 120ns respectively. The intermediate buffer delay is 5ns.

Consider that the system is used for performing 100 tasks.

1. What is the latency of non-pipeline system $t_n = 120+126+121+110+118+120 = 715$ ns
2. What is the latency of pipeline system $t_p = \max(120,126,121,110,118,120)+5 = 131$ ns
3. What is the throughput of pipeline system $= \frac{100}{(6+100-1)131\text{ ns}} = 7270083$ /sec
4. What is the throughput of pipeline system in ideal case $= \frac{1}{131\text{ ns}} = 7633587$ /sec

[NAT]

#Q. The stage delays in a 4-stage pipeline are 60ns, 50ns, 55ns and 80ns. The last stage (with delay 80ns) is replaced with a functionally equivalent design involving two stages with respective delays 60ns and 35ns. The throughput increase of the pipeline is _____ percent?

old

$k = 4$

60, 50, 55, 80

$t_p = 80\ ns$

$\text{throughput} = \frac{1}{t_p}$

$= \frac{1}{80}$

new

$k = 5$

60, 50, 55, 60, 35

$t_p = 60$

$\text{throughput} = \frac{1}{60}$

$\% \text{ of increase} = \frac{\frac{1}{60} - \frac{1}{80}}{\frac{1}{80}} * 100\%$

$= \left(\frac{4}{3} - 1\right) * 100\%$

$= \frac{1}{3} * 100\% = 33.33\%$

[NAT]

#Q. Consider a 6-stage pipeline with delays 2, 4, 3, 5, 3 and 4 cycles. This pipeline is upgraded to a new 8-segment pipeline in which each segment delay is 2 cycle.

1. How much time is saved using new pipeline over old one for 100 tasks? ⇒ 311
2. What is the speed up of new pipeline as compared to old pipeline for 100 tasks?

old :-

$k = 6$

$t_p = 5$

$n = 100$

pipeline time $= (6+100-1)\,5$

$= 525$

new

$k = 8$

$t_p = 2$

pipeline time $= (8+100-1)\,2$

$= 214$

1. Time saved $= 525 - 214 = 311$

2. $= \frac{525}{214} = 2.45$



Topic : Instruction Pipeline

If pipeline processing is implemented on Instruction Cycle



Topic : Instruction Pipeline

Assume 5-stage pipeline

IF: Instruction Fetch
ID: Instruction Decode & Address Calculation
OF: Operand Fetch
EX: Execution
WB: Write Back

Topic : Instruction Pipeline

no. of instns executed (n) = 6

k = 5

no. of cycles = 5 + 6 − 1 = 10

stalls = 3

Total = 13

Clock Cycles

Instructions	1	2	3	4	5	6	7	8	9	10	11	12	13	14
I1	IF	ID	OF	EX	WB									
I2		IF	ID	OF	EX	WB								
I3			IF	ID	OF	EX	WB							
I4				IF	ID	OF	EX	WB						
I5					IF	—	—							
I6														
I7														
I8								IF	ID	OF	EX	WB		
I9									IF	ID	OF	EX	WB	

branch

target

stalls (bubble)



Topic : Instruction Pipeline

no. of instns executed (n) = 9

k = 5

no. of cycles = 5 + 9 − 1 = 13

stalls = 3

Total = 16

Clock Cycles

Instructions	1	2	3	4	5	6	7	8	9	10	11	12	13	14
I1	IF	ID	OF	EX	WB									
I2		IF	ID	OF	EX	WB								
I3			IF	ID	OF	EX	WB							
I4				IF	ID	OF	EX	WB						
I5					IF	-	-	IF	ID	OF	EX	WB		
I6									IF	ID	OF	EX	WB	
I7										IF	ID	OF	EX	WB
I8											IF	ID	OF	EX
I9												IF	ID	OF

WB

EX WB

Condition false

branch



Topic : Instruction Pipeline

In cycle 5 CPU decodes Inst^{n} I4 as branch inst^{n}.

⇓

CPU will know which inst^{n} to execute next, only after condition check. → which happens in execution phase.

Till then pipeline will execute I4 and before it all inst^{ns}; but I5 will be removed from pipeline.

In cycle 7 CPU evaluates condition of branch for inst^{n} I4.

then in cycle 8 CPU fetches next inst^{n} as per branch inst^{n} condition.

stall or stall cycles :-

extra cycles in pipeline due to hazard in pipeline.

Note :-

If branch condition is evaluated in i^{th} stage of pipeline then no. of stalls due to each branch $inst^n = i - 1$

quest^n	Stalls due to each branch inst^n
1) if branch condition is evaluated in 3rd stage of pipeline	3 − 1 = 2
2) If branch target available in 2nd stage of pipeline.	2 − 1 = 1
3) k = 5 if next inst^n after branch inst^n is fetched after complete execution of branch inst^n.	Branch inst^n S1 S2 S3 S4 S5 Next inst^n S1 k − 1 = 5 − 1 = 4

[NAT]

#Q. Consider a program which contains 50 instructions I1, I2 , I3 ..... I50. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I5 and its target is instruction I48. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is 15 ?

No. of inst^{ns} executed $\Rightarrow$ I1 to I5 $\Rightarrow$ 5 $\quad 5-1+1=5$

I48 to I50 $\Rightarrow$ 3 $\quad 50-48+1=3$

$n = 8$

no. of cycles needed w/o stall = $k+n-1 = 5+8-1 = 12$

stalls due to branch inst^n = $4-1 = 3$

Total = $\underline{15}$ Ans.

[NAT]

#Q. In the above question if the pipeline has cycle time 10ns then total time required to execute the program is 150 ns?

$= 15 * 10ns$

$= 150\ ns$

[NAT]

#Q. Consider a program which contains 200 instructions I1, I2 , I3 ..... I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I6 and its target is instruction I16. If during the execution of the program the branch is taken then number of cycles required to execute this program in the given pipeline is 198 ?

$n = 6$ (I1 to I6)

$+(200-16+1)$ (I16 to I200)

$= 191$

w/o hazard $= 5+191-1 = 195$

stalls $= 4-1 = 3$

$= 198$ Ans.

[NAT]

#Q. Consider a program which contains 200 instructions I1, I2 , I3 ..... I200. Further consider a 5-stage pipeline with stages as: Instruction Fetch, Decode, Operand Fetch, Execution and Write-Back. The program contains only 1 branch instruction which is instruction I8 and its target is instruction I108. If during the execution of the program the branch is not taken then number of cycles required to execute this program in the given pipeline is 207 ?

$n = 200$

w/o hazard = $5 + 200 - 1 = 204$

stalls = $4 - 1 = 3$

207 Ans.



2 mins Summary

Topic Pipeline Throughput

Topic Latency

Topic Instruction Pipeline



Happy Learning

THANK - YOU